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Biomechanical Analysis of Disc Golf Throws to Improve Performance

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Introduction

Disc golf (DG) has recently seen a large growth in popularity. UDisc, a DG app reported a 50% increase in their player base in 2020 compared to 2019 [1]. Despite this growth, very little literature correlates technique to performance maximization. This study aimed to gain insight into the biomechanics of forehand and backhand DG throws and to relate these findings to the skill level of the participants. Skill is assessed by relating the throwing mechanics of an athlete to their Professional Disc Golf Association (PDGA) rating, which assigns athletes a number to reflect their performance in PDGA events. The range of PDGA ratings is from the 600s (beginners) to the low 1000s for top professionals [1]. In this study, the participants ranged from the 800s to low 1000 ratings.

Methods

The dataset included 28 participants, including 23 male and 5 female players. Each athlete provided informed consent to participate in the study, and the study was conducted in accordance with the Office of Research Ethics at the University of Waterloo. Subject information was collected and anonymized. The subject was fit with 41 individual motion capture markers spanning the bony landmarks of the subject's shank, thigh, spine, chest, upper arm, forearm, and head. An 8-camera VICON MX20+ motion tracking system (Vicon, Oxford, UK) was used to measure the 3-dimensional kinematics of the throw. Five backhand and five forehand throws were recorded for each participant. The players threw an Innova Disc Golf Destroyer (Innova Disc Golf, Rock Hill, USA) driver into a net, in an indoor lab setting. The participants were instructed to throw with full power while maintaining their standard throwing mechanics, which may have been impacted by the experimental setup (i.e. unfamiliarity with motion capture markers, indoor setting, and no official target).

Once data was collected, motion capture marker points were exported, fit and interpolated to account for gaps in the data. Using the measured data, a musculoskeletal (MSK) model was created in Visual3D (V3D) software. This process began with a static calibration dataset, which was used to match each marker to anatomical landmarks in the software. The defined landmarks were used to create skeletal components, which created a full skeletal model when put together. The articulation of two segments was defined as a joint. Dynamic throwing datasets were applied to the static MSK model, generating a throwing animation for each trial. An automated data pipeline was developed within the software to extract desired kinematics, which were selected from previous literature [2,3]. The selected movements were: elbow flexion/extension, forearm pronation/supination, shoulder horizontal abduction/adduction, and X-Factor (the relative rotation of the trunk to hips). The 3-dimensional coordinates and timesteps were exported from V3D and simulation

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start and end points were specified. The timesteps were then converted into a percentage of the specified swinging motion and the data was plotted and compared in Microsoft Excel. Analysis was conducted on only 6 right-handed subjects, for a total of 13 shots, 4 of which were forehand and 9 of which were backhand.

Results and Discussion

Through the analysis of the plotted kinematics, trends in the higher-rated PDGA players were identified. During a backhand, higher-rated players had faster horizontal abduction of the shoulder, a greater range of shoulder adduction during the wind-up due to contributions of trunk rotation, greater torso range of motion and velocity of trunk rotation, and a greater transition range and velocity of both the wrist pronation/supination and elbow flexion/extension. During a forehand, higher-rated players generated a greater torso contribution due to an increased range of motion and velocity of trunk rotation, a larger degree of elbow extension during the wind-up and most notably, a faster and wider transition range from supination and pronation of the forearm, which generated a prominent wrist snap at release. This can be seen in Figure 1, where generally, higher-rated players (ratings indicated in bracketed numbers) demonstrated the widest range of supination and pronation. Additionally, the steeper transition of said movements indicated a greater wrist snap velocity during disc release.

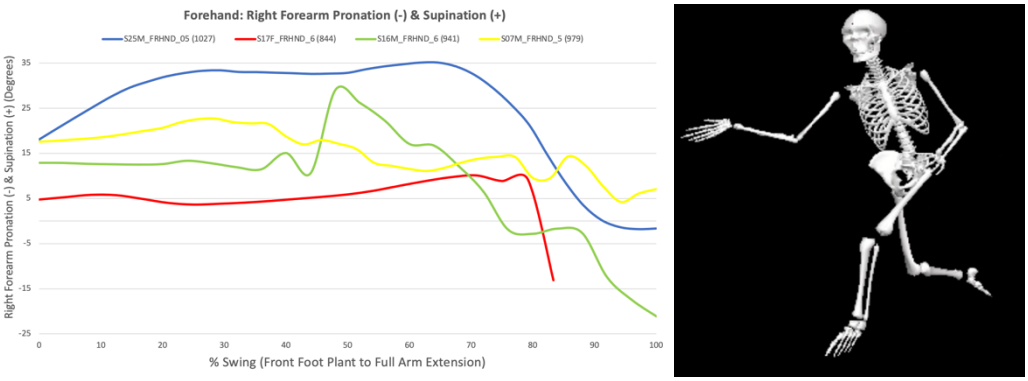

Figure 1: Degree of forearm supination/pronation (left) during forehand throw simulation (right) for variously rated PDGA subjects. The simulation image was captured after disc release, at the beginning of follow-through.

In general, the higher-rated players presented a faster and more seamless transition within the upper-body kinetic chain, while also achieving a greater range of motion for all movement. These biomechanical factors likely contribute to an increased velocity and spin of the disc at launch, which enables the athlete to maximize performance by increasing the carry distance of the projectile during flight [3].The findings of this study will contribute to the growth and quality of the disc golf game by providing both novice and experienced players with biomechanical technique guidelines to improve their performance.

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